Academic year 2023-2024 (Odd Sem)

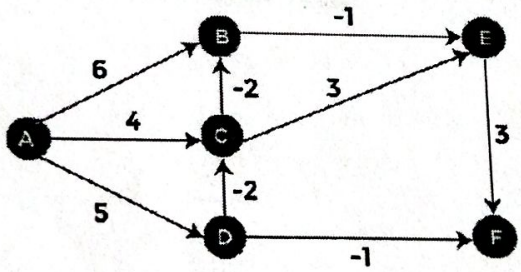
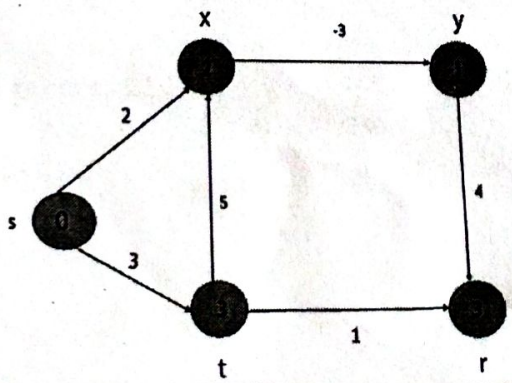
DEPARTMENT OF

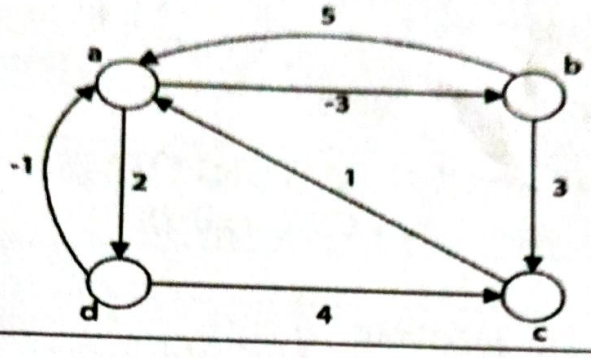
COMPUTER SCIENCE & ENGINEERING

Date	January 2024	Maximum Marks	50
Course Code	21CS55B3	Duration	90 Min
Sem	V Semester	CIE-I	

ADVANCED ALGORITHMS

(Common to CSE and ISE)

Sl. No.	Questions	M	BT	CO
	PART-B			
1a.	Using recursion tree method, solve the recurrence $T(n) = 4T(n/2) + n^2$. Also determine the cost of the entire tree.	6	3	1
1b.	Solve the following recurrence using the Master Theorem, also state which case is applicable. $T(n) = 5T(n/3) + n \log n$	4	3	3
2a.	Illustrate the aggregate method of amortized analysis on the operation INCREMENT in a binary counter.	5	3	2
2b.	Solve the given recurrence relation using substitution method. $T(n) = 2T(n/2) + n \log n$.	5	3	3
3a.	Suppose we perform a sequence of n operations on a data structure in which the ith operation costs i if i is an exact power of 2, and 1 otherwise. Use accounting method of analysis to determine the amortized cost per operation.	5	4	4
3b.	Give the difference between amortized analysis and average-case analysis? Illustrate the potential method for stack operations.	5	2	3
4a.	Demonstrate with an algorithm, the working procedure of bellman ford algorithm for solving the single source shortest path problem considering A as source vertex. 	5	3	3
4b.	Show the execution of Single source shortest paths in DAG for the given graph considering s as a source vertex. 	5	3	3

5.	Use Johnson's algorithm to find the shortest paths between all pairs of vertices in the graph given below. 	10

COURSE OUTCOMES:

Course Outcomes: After completing the course, the students will be able to	
CO1:	Analyze various algorithms for their time and space complexity
CO2:	Demonstrate a familiarity with major algorithms and data structures
CO3:	Apply appropriate design techniques for solving real world problems.
CO4:	Design and implement solutions using appropriate mathematical techniques

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
Total Marks	-	5	40	5	-	-	06	15	24	5	-

****